Chapter 3 & Chapter 4(section 1)

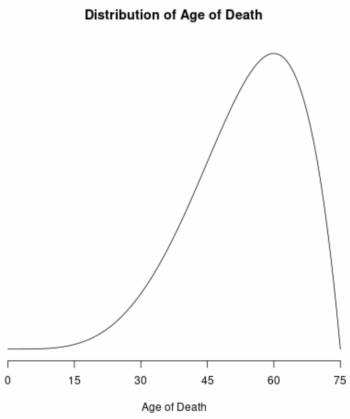
Fred Azizi

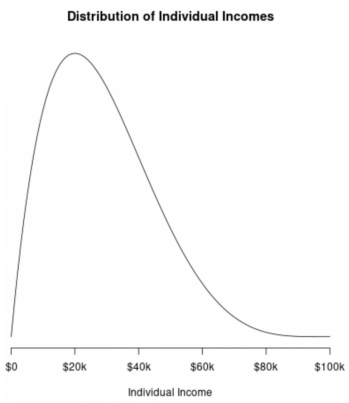
2023-02-21

Quick review

Graphical representations of distribution $\Rightarrow$ Histogram

- Show the frequency distribution for quantitative data over a set of class intervals (Similar to bar chart but works over class intervals).
- Constructed by rectangles whose bases are the intervals and whose heights are the frequencies (or relative frequencies or percent frequencies). **No gaps** between bars.
- We can potentially identify symmetry, distribution skewness, etc.





Quick review (2)

- Time series data are values that correspond to specific measurements taken over a range of time periods.
- Cross-section data are values collected from a number of subjects during a single time period

Figure 1.5 - A Time Series Graph of U.S. Unemployment Rates, 2008–2012

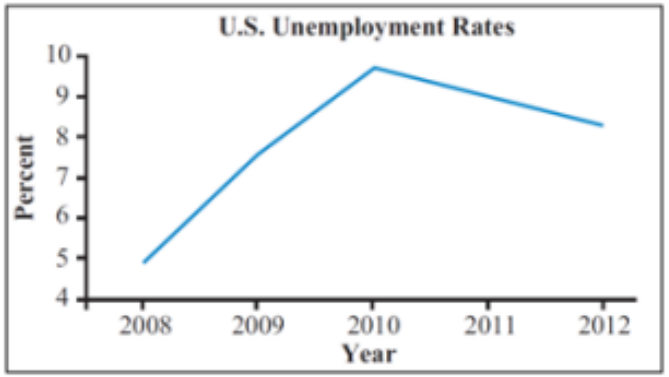
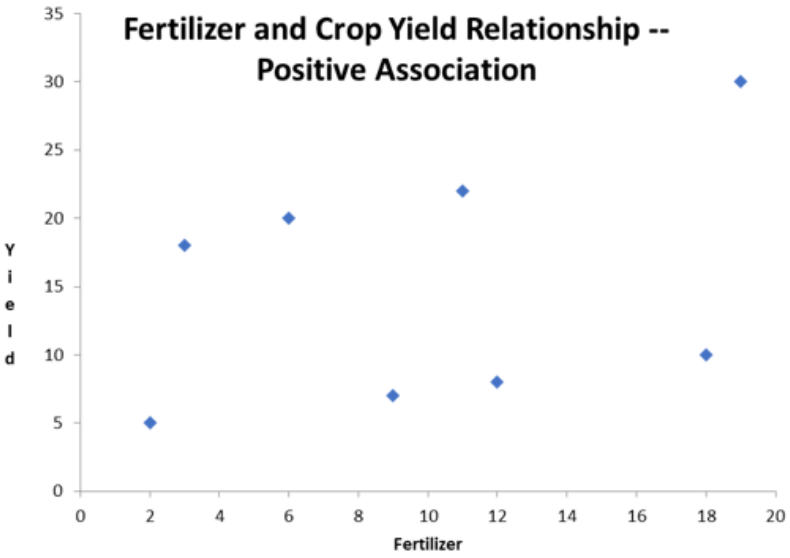
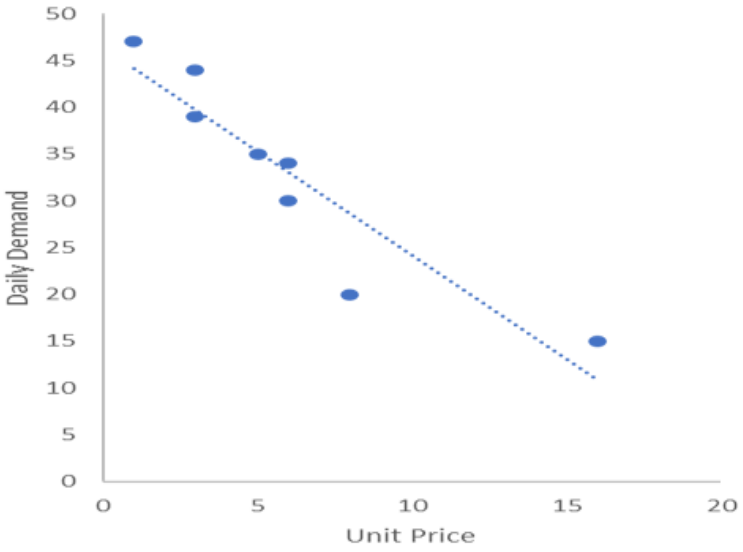


Figure 1.6 - A Cross-Sectional Graph of 2012 Unemployment Rates

Quick review (3)

- Scatter plot: A graph that displays pairs of values as points on a two-dimensional grid.
- The independent/explanatory variable is placed on the horizontal axis, or x-axis. The dependent/response variable is placed on the vertical axis, or y-axis.





Quick review (4)

measure of central tendency:

- Compute the (Arithmetic) Mean to
  - Describe the central location of a single set of interval and ratio data
- Compute the Median to
  - Describe the central location of a single set of interval and ratio or ordinal data
- Compute the Mode to
  - Describe a single set of nominal, ordinal, interval or ratio data